

PaddyGuard AI: A Multimodal AI-Based System for Rice Leaf Disease & Pest Detection, Treatment Recommendation and Knowledge Assistance

Project ID: R26–SE-015

Project Proposal Report
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March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Rice cultivation forms the agricultural backbone of South and Southeast Asia, directly sustaining the livelihoods of more than 350 million smallholder farming families. In Sri Lanka alone, paddy farming covers roughly 770,000 hectares, providing income and subsistence to over 1.8 million households. Despite its central importance, paddy production is severely threatened by foliar diseases, particularly Bacterial Blight, Leaf Blast, and Brown Spot, which can wipe out between 20 and 80 percent of a crop if left untreated. In many rural farming districts, qualified agricultural extension services are limited, leaving farmers to rely on guesswork, delayed consultation, or inappropriate agrochemical use when disease symptoms first appear.

Although automated image-based plant disease detection has made significant progress in recent years, deployed systems still exhibit an important limitation: they often produce highly confident but incorrect predictions when given images that fall outside the distribution of their training data. Farmers may photograph soil, weeds, insects, or plants affected by previously unseen diseases, yet traditional classifiers still force these inputs into one of the known disease categories. This can lead to unreliable predictions and incorrect treatment decisions.

This research designs, implements, and evaluates Component 2 of the PaddyGuard AI system: an explainable rice leaf disease classification module with integrated out-of-distribution (OOD) detection capability. Three convolutional neural network architectures ResNet50, EfficientNetB3, and DenseNet121 are fine-tuned using ImageNet pre-trained weights on a multi-source rice leaf image dataset containing four target classes. Three complementary OOD detection methods ODIN, Mahalanobis Distance, and OpenMax are integrated and evaluated using a probe dataset consisting of non-rice leaves and unseen disease images. In addition, Grad-CAM visualizations are generated to highlight disease-relevant regions that influence model predictions, improving interpretability and user trust.

The proposed system aims to provide reliable and transparent disease diagnosis while maintaining high classification accuracy and efficient inference performance suitable for mobile deployment. By supporting early disease identification for farmers, the system contributes to precision agriculture and aligns with Sustainable Development Goal 2 (Zero Hunger) and SDG 9 (Industry, Innovation, and Infrastructure).

Keywords: Rice leaf disease detection, Out-of-distribution detection, Convolutional neural networks, Transfer Learning, Precision agriculture

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my supervisor, Dr. Nathali Silva, of the Sri Lanka Institute of Information Technology, for her expert technical guidance, timely feedback, and sustained encouragement throughout every stage of this proposal. Her deep understanding of machine learning systems and her patience in working through the methodological challenges of this research have been invaluable.

I would also like to thank my co-supervisor, Ms. Poorna Panduwawala, whose constructive observations on the experimental design and system architecture consistently pushed the work towards greater rigor and clarity. My sincere thanks also go to Mrs. Anoma Gunarathna, external field expert, whose extensive knowledge of rice cultivation practices in Sri Lanka ensured that this research remains grounded in the realities of farm-level deployment rather than controlled laboratory conditions. I also thank the Department of Agriculture of Sri Lanka for providing domain guidance on the locally prevalent disease presentations of Bacterial Blight, Leaf Blast, and Brown Spot.

I am grateful to my project group colleagues for the collaborative technical discussions that shaped the broader PaddyGuard AI architecture, and to the farming communities of the Northwestern Province who participated in early user consultations. Their feedback directly informed usability requirements of this system. Finally, I thank my family for their patience and unwavering moral support throughout this research journey.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DL	Deep Learning
FAISS	Facebook AI Similarity Search
Grad-CAM	Gradient-weighted Class Activation Mapping
ML	Machine Learning
OOD	Out-of-Distribution
ODIN	Out-of-Distribution Detector for Neural Networks
RAG	Retrieval-Augmented Generation
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
TTA	Test-Time Augmentation

1. INTRODUCTION

1.1 Background and Context

Rice (*Oryza sativa* L.) is the staple food for more than half of the world's population and remains one of the most important agricultural crops across South and Southeast Asia. In Sri Lanka, paddy cultivation occupies a significant portion of agricultural land and provides livelihoods for a large number of farming families. As a result, maintaining stable rice production is essential for both food security and the national economy [1]. However, rice cultivation is highly vulnerable to biological threats, particularly leaf diseases that can significantly reduce crop yield and quality if they are not detected at an early stage [2].

Among the most common and economically damaging rice leaf diseases are Bacterial Blight, Leaf Blast, and Brown Spot. Bacterial Blight, caused by *Xanthomonas oryzae* pv. *Oryzae*, leads to yellowing and wilting of rice leaves, particularly under humid environmental conditions. Leaf Blast, caused by *Magnaporthe oryzae*, is considered one of the most destructive rice diseases worldwide and can cause severe damage to plant tissues and grain production. Brown Spot, caused by *Cochliobolus miyabeanus*, produces dark lesions on rice leaves and is often associated with poor soil conditions and drought stress. Under favorable conditions, these diseases can cause significant yield losses if they are not identified and treated promptly [3].

Traditionally, disease diagnosis relies on visual inspection by agricultural officers or expert consultation. However, in many rural farming areas, access to agricultural experts is limited. As a result, farmers often rely on personal judgment or delayed treatment decisions, which can lead to incorrect pesticide use, increased production costs, and potential environmental damage [4]. These limitations highlight the need for accessible and reliable technological solutions that can assist farmers in identifying rice diseases directly in the field.

Recent advancements in artificial intelligence and deep learning have created new opportunities for automated plant disease detection. In particular, convolutional neural networks (CNNs) have demonstrated strong performance in image-based plant disease classification. Despite these advances, many existing systems operate under a closed-world assumption, meaning that every input image is expected to belong to one of the predefined disease categories used during model training. In real agricultural environments, this assumption is often unrealistic. Farmers may upload images containing healthy leaves, weeds, soil backgrounds, insects, blurred images, or previously unseen disease conditions. In such cases, traditional classifiers may produce highly confident but incorrect predictions, which can lead to inappropriate treatment recommendations [5].

To address these challenges, this research proposes the development of an explainable rice leaf disease classification module with integrated Out-of-Distribution (OOD) detection. This module forms Component 2 of the PaddyGuard AI system, a multimodal agricultural support platform designed to assist farmers with disease detection and crop management. By combining deep learning classification with OOD detection and visual explainability, the proposed system aims to improve the reliability, transparency, and practical usability of automated disease diagnosis tools for real-world agricultural environments. The research also contributes to Sustainable

Development Goal 2 (Zero Hunger) by supporting food security and SDG 9 (Industry, Innovation and Infrastructure) through the development of innovative agricultural technologies.

1.2 Literature Review

During the last decade, the scientific community has conducted many studies on the application of deep learning for plant and rice disease classification. The following review summarizes several key research contributions in this area and highlights the remaining limitations that motivate the proposed work.

Mohanty et al. [6] presented one of the earliest large-scale studies on plant disease classification using deep learning. Their work used a dataset containing 54,306 images across 26 diseases from 14 crop species and achieved an accuracy of 99.35% using a fine-tuned AlexNet model. This study demonstrated the potential of deep learning for automated plant disease detection. However, the experiments were conducted under controlled laboratory conditions, and the results may not generalize well to real agricultural environments where lighting conditions, occlusions, and background noise vary significantly.

Lu et al. [7] developed a convolutional neural network specifically designed for rice disease detection and achieved an accuracy of 94.7% when classifying four rice diseases. Their work showed the benefits of using domain-specific datasets and transfer learning techniques. However, the system assumed that every input image belonged to one of the predefined disease categories and did not include a mechanism to reject unknown or non-rice images.

Chen et al. [8] investigated the use of EfficientNet architectures for agricultural disease classification. Their results showed that EfficientNetB3 and EfficientNetB4 could achieve comparable or better accuracy than ResNet-based models while using fewer parameters. This finding influenced the selection of EfficientNetB3 as one of the candidate architectures evaluated in the present study. However, the evaluation in that work focused only on in-distribution classification accuracy and did not examine system behavior when presented with out-of-distribution inputs.

Ferentinos [9] evaluated multiple convolutional neural network architectures on the PlantVillage dataset and reported an accuracy of 99.53% for plant disease classification. Nevertheless, the study also noted that performance significantly decreased when models were tested on real field images. This observation highlights the importance of data augmentation and dataset diversity when designing practical disease detection systems.

Liang et al. [10] introduced the ODIN (Out-of-Distribution detector for Neural networks) method for detecting inputs that do not belong to the training distribution. ODIN modifies the softmax confidence score using temperature scaling and small input perturbations to better distinguish between in-distribution and out-of-distribution samples. The method demonstrated strong performance on multiple benchmark datasets without requiring retraining of the base classifier.

Lee et al. [11] proposed a Mahalanobis distance-based approach for out-distribution detection using feature representations extracted from intermediate layers of deep neural networks. Their method models class-conditional Gaussian distributions in feature space and computes a confidence score based on the distance between a test sample and the nearest class meaning. The

method showed strong detection performance, particularly for high-dimensional image embeddings.

Bendale and Boulton [12] proposed OpenMax as an alternative to the traditional SoftMax classifier for open-set recognition. OpenMax applies Extreme Value Theory to estimate the probability that a sample belongs to an unknown class. This probabilistic approach allows models to reject unfamiliar inputs instead of forcing them into known categories.

Selvaraju et al. [13] introduced Grad-CAM, a visualization technique that highlights the image regions that contribute most strongly to a model's prediction. Grad-CAM has been widely used to improve interpretability in deep learning systems. In the context of plant disease detection, Grad-CAM can help verify whether the model focuses on actual disease lesions rather than irrelevant background features.

Several commercial tools for plant disease diagnosis already exist, including the Plantix mobile application and the IRRI Rice Doctor tool. While these tools provide image-based disease diagnosis, they generally rely on closed-set classifiers and do not include mechanisms for detecting unknown inputs or providing visual explanations for predictions. Bock et al. [14] reported that some commercial tools produced incorrect high-confidence predictions when evaluated on atypical disease samples. Similarly, other systems such as RICEPEST and Paddy Doctor lack OOD detection and explainability mechanisms [15].

These limitations highlight the need for disease classification systems that combine accurate prediction with reliable rejection of unknown inputs and improved model transparency.

1.3 Research Gap

A review of the existing literature reveals several important limitations in current rice leaf disease detection systems, which motivate the need for further research.

The first limitation is that most existing deep learning-based disease classification systems operate under a closed-world assumption, where every input image is expected to belong to one of the disease classes used during model training. In real agricultural environments, this assumption is unrealistic. Farmers may upload images that contain healthy leaves, non-rice plants, soil backgrounds, insects, blurred images, or previously unseen disease conditions. When such inputs are provided, conventional classifiers may still produce highly confident predictions, even when the prediction is incorrect. This can lead to misleading diagnoses and potentially harmful crop management decisions.

The second limitation is the lack of systematic integration and evaluation of Out-of-Distribution (OOD) detection techniques in rice disease classification systems. Although several OOD detection methods such as ODIN, Mahalanobis distance, and OpenMax have been studied in the broader machine learning domain, their effectiveness has not been thoroughly evaluated within rice leaf disease classification frameworks. Issues such as threshold calibration, false rejection rates for genuine disease images, and computational performance in practical deployment environments require further investigation.

The third limitation relates to model interpretability and user trust. Many existing AI-based plant disease detection systems provide only a predicted class label without any explanation of how the decision was made. For farmers and agricultural officers, especially those with limited technical knowledge, this lack of transparency makes it difficult to determine whether the prediction is based on actual disease symptoms or irrelevant background patterns. As a result, the adoption of AI-based decision-support tools in agriculture may remain limited.

Therefore, this research aims to address these limitations by developing a rice leaf disease classification system that integrates Out-of-Distribution detection and visual explainability. The proposed system will evaluate multiple OOD detection methods and incorporate Grad-CAM visual explanations to highlight disease regions in leaf images. By combining accurate disease classification with reliable unknown input rejection and interpretable predictions, the system aims to improve the reliability, transparency, and practical usability of AI-based disease diagnosis in real-world paddy farming environments.

2. OBJECTIVES

2.1 Main Objective

To design, develop, and evaluate an explainable deep learning-based rice leaf disease classification system that can accurately identify Bacterial Blight, Leaf Blast, Brown Spot, and Healthy leaves from uploaded images, while also detecting and rejecting out-of-distribution (OOD) inputs that do not belong to the known classes.

2.2 Specific Objectives

- To collect, clean, and augment a diverse dataset of rice leaf images representing the four target classes, together with additional OOD samples such as non-rice plants, soil, water backgrounds, and unseen disease categories.
- To implement and compare transfer learning-based CNN architectures, namely ResNet50, EfficientNetB3, and DenseNet121, using classification accuracy, macro F1-score, and inference efficiency as model selection criteria.
- To integrate and evaluate ODIN, Mahalanobis Distance, and OpenMax for OOD detection and determine suitable thresholds for practical deployment.
- To generate Grad-CAM visual explanations and assess whether the model focuses on biologically meaningful disease regions.
- To export the final model into a lightweight deployment format suitable for server-side, mobile, or edge-based inference.
- To achieve a target in-distribution classification accuracy above 92% and an OOD rejection rate above 85% on the final evaluation set.

3. METHODOLOGY

3.1 Research Approach

This study follows the Design Science Research (DSR) approach, which is suitable for computing research that focuses on designing and evaluating a practical software artefact [16]. The research will be carried out through iterative stages of design, implementation, testing, and refinement. In each cycle, the classification model, OOD detection methods, and explainability pipeline will be improved based on experimental results. This approach helps ensure that the final system is both technically sound and practically useful in real-world agricultural environments.

3.2 Dataset Preparation

The dataset will be prepared using multiple sources in order to improve generalization and make the system more robust in field conditions. The main in-distribution dataset will contain rice leaf images from publicly available sources such as Kaggle and other plant disease image repositories. Where possible, the dataset will also be supplemented with locally collected field images representing Sri Lankan paddy farming conditions. The target classes will be:

- Bacterial Blight
- Leaf Blast
- Brown Spot
- Healthy Leaf

To support OOD detection, additional images will be collected from non-rice leaves, weeds, soil, water backgrounds, insects, and visually similar but previously unseen disease conditions. All images will undergo preprocessing steps such as resizing, normalization, and quality checking. Data augmentation methods such as horizontal flipping, rotation, brightness adjustment, contrast variation, and blur will be applied to improve robustness under real-world conditions. The dataset will be divided into training, validation, and testing sets using a 70:15:15 split while maintaining class balance.

3.3 Model Architecture and Training

This study will evaluate three CNN architectures: ResNet50, EfficientNetB3, and DenseNet121. These models have been chosen because they are widely used in image classification tasks and offer a balance between predictive performance and computational efficiency. Each model will be initialized with ImageNet pre-trained weights and fine-tuned on the rice leaf dataset. The final classification layer will be modified to produce four output classes.

Training will be carried out using the Adam optimizer with an initial learning rate of 1×10^{-4} . Cross-entropy loss will be used as the loss function, while weight decay and early stopping will

be applied to reduce overfitting. Model selection will be based on validation accuracy, macro F1-score, parameter efficiency, and inference speed.

Table 1: Comparison of CNN Architectures for Disease Classification

Architecture	Parameters (M)	Top 1 Acc. (ImageNet)	Depth	Reason for Selection
ResNet50	25.6	76.1%	50	Strong baseline, widely validated
EfficientNetB3	12.2	81.6%	30	High accuracy with fewer params
DenseNet121	8.0	74.9%	121	Feature reuse; good for small datasets

3.4 OOD Detection Integration

To improve reliability in real-world use, three OOD detection methods will be integrated with the selected classifier. ODIN improves OOD detection by using temperature scaling and small input perturbations to enlarge the confidence gap between in-distribution and OOD samples.

Mahalanobis Distance uses class-conditional feature distributions to measure how closely a test image matches known class patterns in feature space. OpenMax replaces the standard SoftMax output with a calibrated mechanism that estimates the probability that a sample belongs to an unknown class.

Thresholds for each method will be tuned on the validation set to balance sensitivity and rejection performance. Comparative evaluations will be carried out using AUROC, AUPR, and FPR95

Table 2: OOD Detection Method Comparison

Method	Requires Retraining	Computational Overhead	Key Strength
ODIN	No	Low	Simple, training-free baseline
Mahalanobis Distance	No	Medium	Uses deep feature statistics
OpenMax	Partial (calibration)	Medium	Probabilistic open-set modelling

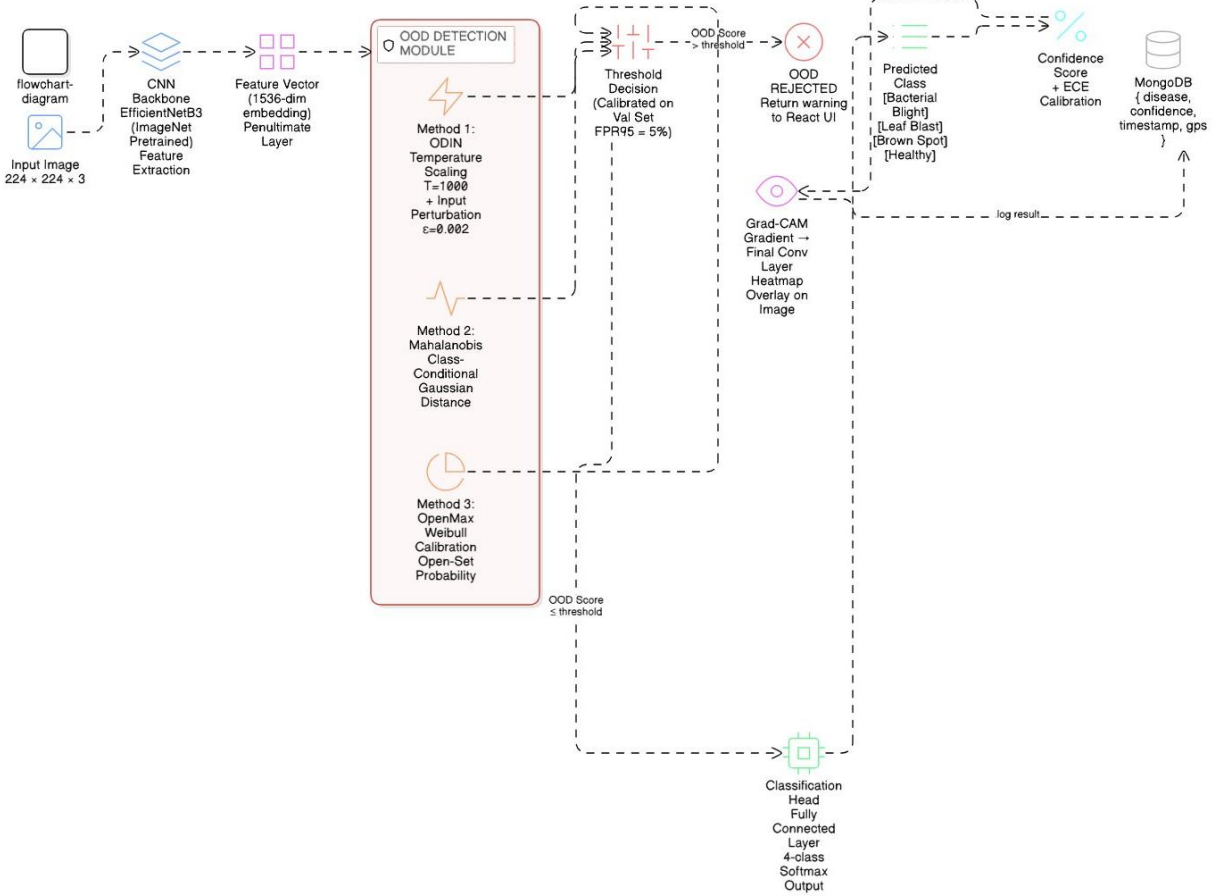


Figure 1: CNN Pipeline with OOD Detection Integration

3.5 Explainability

To improve interpretability, Grad-CAM will be applied to the final convolutional layers of the selected model. This will generate heatmaps that show which image regions most influenced the prediction. These visual explanations will be displayed together with the predicted class and confidence score. A qualitative review by an agricultural domain expert will be used to assess whether the highlighted regions correspond to meaningful lesion areas rather than irrelevant background features.

3.6 Evaluation Metrics

The classification performance of the system will be evaluated using:

- Accuracy
- Precision
- Recall
- Macro F1-score

OOD detection performance will be evaluated using:

- AUROC
- AUPR
- FPR95

In addition, inference latency will be measured to assess deployment suitability, and Expected Calibration Error (ECE) will be used to assess model confidence calibration

3.7 Ethical Considerations

This research will use publicly available datasets and, where applicable, locally collected field images obtained with farmer consent. No personally identifiable information will be collected as part of the image dataset. All stored data will be securely managed, and access will be restricted to authorized project members. The study does not involve clinical or interventional human experimentation.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall System Overview

PaddyGuard AI is designed as a multimodal agricultural support platform consisting of four main intelligent components:

1. Voice-based disease diagnosis
2. Rice leaf disease classification with OOD detection
3. Treatment recommendation and advisory support
4. Paddy pest detection

These components share a common frontend and backend environment while functioning as separate intelligent modules.

This proposal focuses on Component 2, which receives rice leaf images from the user interface, performs disease classification, applies OOD screening, and generates explainability outputs. The result is then forwarded to the treatment recommendation component for further support.

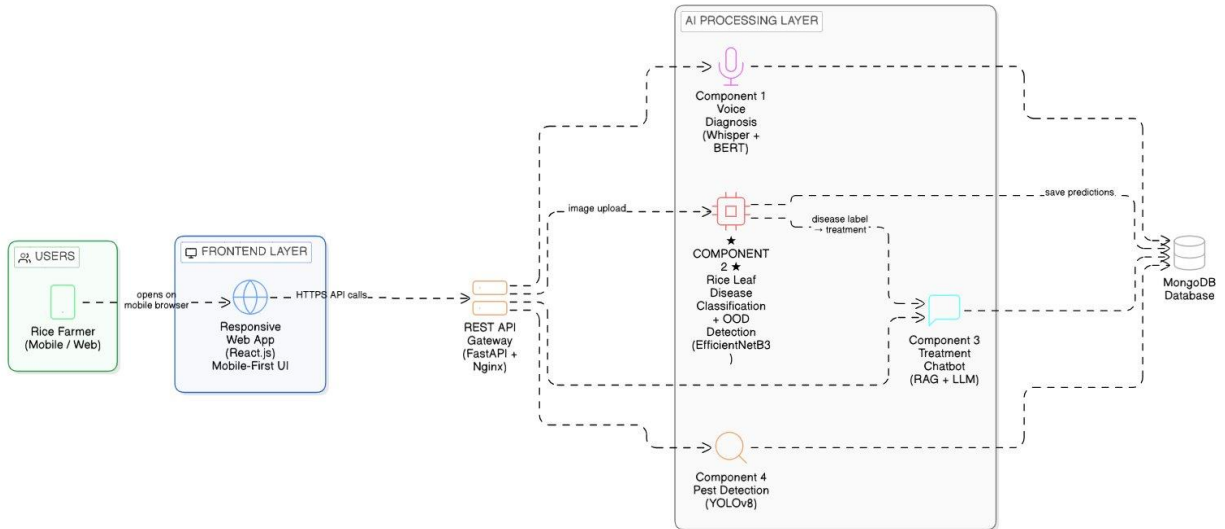


Figure 2: High-Level System Architecture of PaddyGuard AI

High-Level Architecture of Component 2: Rice Leaf Disease Classification with Out-of-Distribution Detection

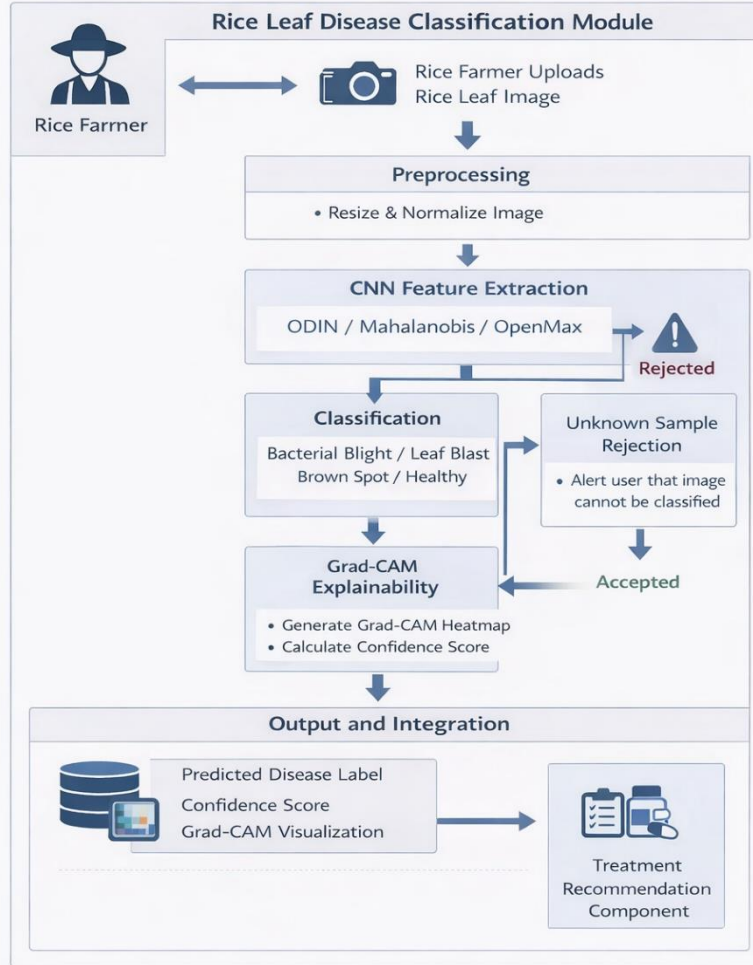


Figure 3: High-Level System Architecture of Component 2

4.2 Component 2 Internal Data Flow

The internal workflow of Component 2 consists of five main stages. First, the uploaded input image is resized and normalized. Second, a backbone CNN extracts deep feature representations from the image.

Third, the OOD detection module evaluates whether the sample belongs to the known class distribution. If the image is rejected as OOD, a user-friendly rejection message is returned. If the image is accepted, it proceeds to the classification stage, where it is classified into one of the four disease categories.

Finally, Grad-CAM heatmaps and confidence scores are generated and packaged for display in the frontend.

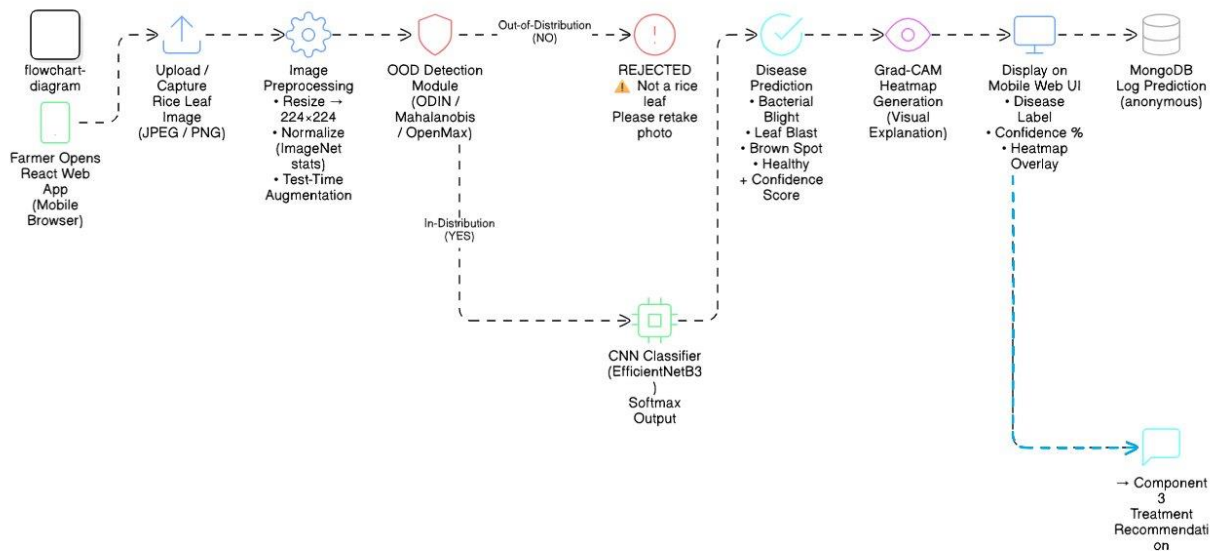


Figure 4: Component 2 — Disease Classification Module Data Flow

4.3 Scalability and Security Considerations

The final model will be exportable in formats suitable for both server-side and on-device inference, such as ONNX and TensorFlow Lite. The backend API will be developed using FastAPI and deployed in a containerized environment for scalability. All communication between the mobile client and the server will be protected using TLS 1.3. By default, uploaded images will not be permanently stored unless explicitly required for research improvement with user consent. Only anonymized prediction logs will be retained for monitoring and future retraining.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Primary Users: Smallholder rice farmers in Sri Lanka and other rice-growing regions, particularly those with limited formal education and technology literacy.
- Secondary Users: Agricultural extension officers and agronomists who may use the system for rapid field-level diagnosis.
- Tertiary Stakeholders: Agricultural research institutions, government ministries of agriculture, and NGOs involved in food security programmes.
- System Administrators: IT personnel responsible for model monitoring, retraining, and platform maintenance.

5.2 Functional Requirements

Table 3: Functional Requirements

ID	Requirement	Description
FR01	Image Capture	The system shall accept rice leaf images from the mobile device camera or gallery in JPEG/PNG format.
FR02	Image Preprocessing	The system shall automatically resize, normalize, and optionally augment input images before inference.
FR03	Disease Classification	The system classifies valid input images into one of four categories: Bacterial Blight, Leaf Blast, Brown Spot, or Healthy.
FR04	OOD Detection	The system shall detect and reject images that do not belong to any of the four training categories, returning a user-friendly rejection message.
FR05	Confidence Display	The system shall display the classification confidence score alongside the predicted disease label.
FR06	Explainability Map	The system shall generate and display a Grad-CAM heatmap overlaid on the original image for each in-distribution prediction.
FR07	Result Logging	The system shall anonymously log prediction outcomes for ongoing model monitoring with user consent.
FR08	API Integration	The classification API shall expose a RESTful endpoint consumable by the PaddyGuard AI frontend and by the Treatment Recommendation module (Component 3).

5.3 Non-Functional Requirements

Table 4: Non-Functional Requirements

ID	Category	Requirement
NFR01	Performance	The system shall complete inference and return a result within 500 milliseconds under standard network conditions.
NFR02	Accuracy	The system shall achieve a minimum classification accuracy of 92% on the held-out test set.
NFR03	OOD Rejection Rate	The OOD detector shall correctly reject at least 85% of out-of-distribution probe samples at a calibrated threshold.
NFR04	Usability	The user interface shall be operable by users with low digital literacy, featuring iconographic guidance and multilingual support (Sinhala, Tamil, English).
NFR05	Reliability	The system shall maintain 99% uptime during peak agricultural seasons.
NFR06	Security	All image data transmitted to the server shall be encrypted using TLS 1.3.
NFR07	Scalability	The backend shall support horizontal scaling to accommodate up to 10,000 concurrent inference requests.
NFR08	Portability	The model shall be exportable in TensorFlow Lite format for on-device inference on Android devices with Android 9.0 or higher.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Our main target market is smallholder rice farmers of Sri Lanka, India, Bangladesh, Vietnam and Indonesia these countries contribute to more than 65% of the worlds rice production. Our typical customer persona is a male farmer between 35 and 55 years old and manage 1 to 5 acres of paddy, has of a low level of education and owns a basic Android mobile phone. Other market segments include farmers cooperation, agricultural extension offices and Agri-input companies (i.e. fertilizer or pesticide companies) that are looking for agent for farmers.

6.2 Value Proposition

Paddy Guard's A.I. Component 2 offers three distinct value propositions.¹ First, reduces the time to diagnosis for disease from days (when waiting for an extension officer to arrive) to seconds (on-device A.I. inference).²O.O.D detection feature also sharpens PaddyGuard AI's value proposition away from competitors by avoiding misdiagnosis or misapplication that could be caused by nonstandard/orienting images.³And the Grad-CAM feature helps build trust in AI through visual validation of predictions made on leaf symptoms.

6.3 Revenue Model

The commercialization approach is based on a freemium SaaS model. The first tier provides a limited number of free monthly queries for disease diagnosis and constitutes a farmer acquisition approach. The second tier, directed at cooperative and extension services, offers limitless queries, logging of all diagnoses, multi-field management capabilities and an API for integration with farm management software. An enterprise tier is offered to government agricultural organizations and NGOs, which offers bulk licensing, re-training of models on local disease presentations, and access to own cloud hosting.

6.4 Pricing Strategy

Indicative pricing: Basic free (limited to 20 diagnoses per month per user). Premium LKR 990/month or USD 3.00/month per individual farmer. Cooperative /Enterprise Negotiated depending on total number of users and number of API calls, US\$ 150–500/month per organization.

6.5 Competitive Advantage

Our main comparative advantage against the current solutions (Pantex, Rice Doctor, Paddy Doctor) is the inclusion of OOD detection, a feature not available from any competitors at the time of this writing. Other advantages are multi-language support for Sri Lankan languages, dataset curation of local disease images and close integration with other ideal PaddyGuard AI components (voice diagnosis, pest detection and treatment recommendation). Together these from a high switching cost and complete value stack that no single competitor can match unless it invests equivalent in research.

6.6 Intellectual Property

The trained model weights, OOD calibration parameters and Sri Lankan field image dataset comprise the core intellectual property assets. The research team will investigate patent applications for the OOD-integrated disease detection pipeline with the Sri Lanka National Intellectual Property Office (NIPO). The codebase will be released under a limited, academic-use-only license which allows for non-commercial research use, while maintaining the project's commercial rights with the project team.

7. BUDGET AND JUSTIFICATION

Table 5: Budget Breakdown

Item #	Item	Estimated Cost (LKR)	Justification
1	Google Colab Pro (6 months)	18,000	GPU access for model training (T4/A100). Essential for deep learning experimentation.
2	Kaggle Pro (Annual)	0	Free tier sufficient; datasets accessed from Kaggle public repository.
3	Cloud Storage (AWS S3 / GCS)	6,000	Dataset storage, model checkpoints, and API hosting during prototype phase.
4	Domain & SSL Certificate	3,500	Required for secure HTTPS API endpoint for prototype deployment.
5	Field Data Collection Travel	15,000	Site visits to Polonnaruwa and Kurunegala for field image collection.
6	Agricultural Expert Consultation	10,000	Domain expert review of Grad-CAM outputs and dataset annotation verification.
7	Software Licenses	0	All core tools (PyTorch, FastAPI, Docker) are open-source.
8	Contingency (10%)	5,250	Buffer for unforeseen GPU overruns or additional data acquisition needs.
	TOTAL	57,750 LKR (~USD 175)	

8. WORK BREAKDOWN STRUCTURE (WBS)

Work Breakdown Structure For Rice Leaf Disease Classification with Out-of-Distribution Detection using Deep Learning

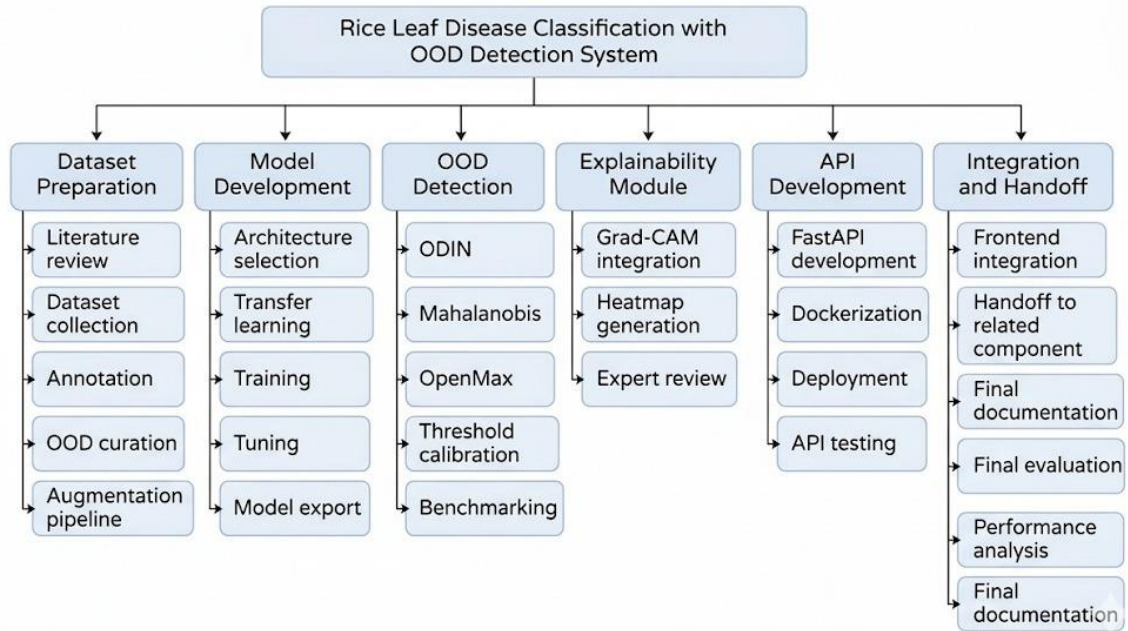


Figure 5: Work Breakdown Structure — Component 2

9. GANTT CHART

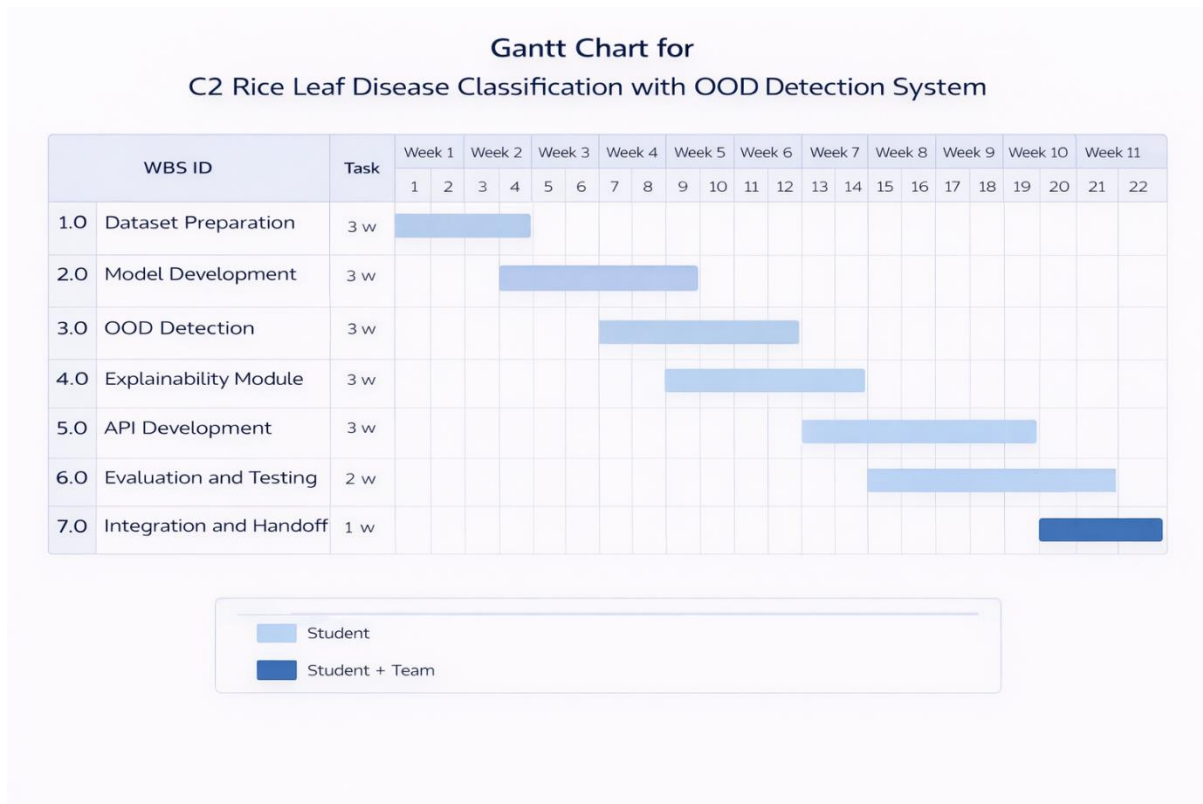


Figure 6: Gantt Chart — Project Timeline

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- Mohanty et al. (2016) — <https://doi.org/10.3389/fpls.2016.01419> (Open Access, Frontiers)
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- Mahalanobis OOD (Lee et al., 2018) — <https://arxiv.org/abs/1807.03888> (Free PDF on arXiv)
- OpenMax (Bendale & Boulton, 2016) — <https://ieeexplore.ieee.org/document/7780542> (IEEE — use SLIIT library proxy)
- Grad-CAM (Selvaraju et al., 2017) — <https://arxiv.org/abs/1610.02391> (Free PDF on arXiv)
- EfficientNet (Tan & Le, 2019) — <https://arxiv.org/abs/1905.11946> (Free PDF on arXiv)
- Design Science Research (Hevner et al., 2004) — <https://doi.org/10.2307/25148625> (Access via JSTOR / SLIIT library)
- Savary et al. (2019) — <https://doi.org/10.1038/s41559-018-0793-y> (Nature — use SLIIT library proxy)